ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/384443385

Practical Application of Linear programming technique in textile spinning industry

Article · September 2024

CITATIONS
0

READS
62

1 author:



Sujai Balasubramaniam
Inarco Private Limited
15 PUBLICATIONS **0** CITATIONS

SEE PROFILE

All content following this page was uploaded by Sujai Balasubramaniam on 29 September 2024.

The user has requested enhancement of the downloaded file.

Practical Application of Linear programming technique in textile spinning industry



Sujai Balasubramaniam
Digitally signed by Sujai Balasubramaniam
DN: C=IN, OU="Inarco Pvt Ltd ", O=INARCO, CN=Sujai Balasubramaniam, E=products@inarco.com
Reason: I am the author of this document
Location: coimbatore
Date: 2024.09.04 20:18:11+05'30'
Foxit PDF Editor Version: 11.0.1

SUJAI BALASUBRAMANIAM

Deputy General Manager – Application Technology
Inarco Private Limited
Email: products@inarco.com

Abstract:

This article deals about the practical application of Linear programming technique in textile spinning industry so that optimum products can be selected based on their discrete parameters, Maximization of profit and minimization of waste which can be done by choosing ideal raw material for a given count ,etc.

Key words:

Linear programing, spinning machine, rubber cots, metallic card clothing, metallic rings, Traveler, raw material

Linear programing:

Linear programming (LP; also called linear optimization) is a method to achieve the best outcome (such as maximum profit or lowest cost) in a mathematical model whose requirements are represented by linear relationships. Linear programming is a special case of mathematical programming (mathematical optimization).

Mathematical technique used in computer modeling (simulation) to find the best possible solution in allocating limited resources (energy, machines, materials, money, personnel, space, time, etc.) to achieve maximum profit or minimum cost. However, it is applicable only where all relationships are linear, and can accommodate only a limited class of cost functions. For problems involving more complex cost functions, another technique called 'mixed integer modeling' is employed.

Standard form

Standard form is the usual and most intuitive form of describing a linear programming problem. It consists of the following three parts:

- A **linear function to be maximized**

  e.g. $f(x_1, x_2) = c_1 x_1 + c_2 x_2$

  - **Problem constraints** of the following form

    $$a_{11} x_1 + a_{12} x_2 \leq b_1$$
    $$a_{21} x_1 + a_{22} x_2 \leq b_2$$
    $$a_{31} x_1 + a_{32} x_2 \leq b_3$$

    - **Non-negative variables**

    e.g.

    $$x_1 \geq 0$$
    $$x_2 \geq 0$$

    The problem is usually expressed in *matrix form*, and then becomes:

    $$\max\{c^{\mathrm{T}} x \mid Ax \leq b \wedge x \geq 0\}$$

Other forms, such as minimization problems, problems with constraints on alternative forms, as well as problems involving negative variables can always be rewritten into an equivalent problem in standard form.

The simplex method is an algorithm used to solve linear programming (LP) problems. It is designed to find the optimal solution to a problem where the objective is to maximize or minimize a linear function, subject to a set of linear constraints.

Key Concepts of the Simplex Method

1. **Linear Programming Problem**:
   Objective Function: This is the function you want to maximize or minimize. It is usually in the form of $Z = c_1 x_1 + c_2 x_2 + \cdots + c_n x_n$ where $c_1, c_2, \ldots, c_n$ are coefficients, and $x_1, x_2, \ldots x_n$, are decision variables.
2. **Constraints**: These are the linear inequalities or equalities that restrict the values of the decision variables. They are generally in the form $a_{11} x_1 + a_{12} x_2 + \cdots + a_{1n} x_n \leq b_1$ for the first constraint, and so on.
3. **Non-negativity Constraints**: The decision variables $x_1, x_2, \ldots x_n$ are usually required to be non-negative, meaning $x_i \geq 0$

Steps of the Simplex Method:

1. Convert to Standard Form:

- If the problem involves inequalities, convert them into equalities by adding slack variables. For example, the constraint $x_1 + 2x_2 \leq 8$ can be rewritten as $x_1 + 2x_2 + s_1 \leq 8$, where s1≥0 is a slack variable.

2. Set Up the Initial Simplex Representation:

- The tableau is a tabular representation of the linear programming problem, including the objective function, constraints, and slack variables.

3. Identify the Pivot Element:

- The pivot element is used to perform row operations to improve the solution. The pivot column is chosen based on the most negative coefficient in the objective function row (indicating the direction of improvement). The pivot row is determined by dividing the right-hand side values by the corresponding positive values in the pivot column, selecting the smallest ratio.

4. Perform Row Operations:

- Use the pivot element to adjust the tableau so that the new solution moves towards the optimal point. This involves making the pivot element equal to 1 and adjusting other elements in the pivot column to 0 by appropriate row operations.

5. Repeat the Process:

- Continue identifying new pivot elements and performing row operations until there are no more negative coefficients in the objective function row. This indicates that the current solution is optimal.

6. Read the Solution:

- The optimal values of the decision variables can be read directly from the final tableau. The value of the objective function is given by the right-hand side value in the objective function row.

Example # 1:

Costing of Raw material:

Raw material cost is a big factory for any textile spinning industry, it nearly accounts for 60% of the total production cot yet selection of raw material based on its parameter is very vital for a spinners. In this case we are going to deal with the optimum mixing of 3 different variety of fibres for producing cotton yarn of 90s Ne

Properties of Raw Materials that affect the final product:

- Length
- Strength
- Maturity Coefficient
- Fineness – Micronaire

Here LPT can be used to minimize total cost of raw material and Maximize quality of output yarn.

Parameters:

Sr.No	Parameter	Min Norms required by the mills to produce 120s Ne
1	Fibre length in mm	36 mm
2	Fibre strength	23 Gms per tex
3	Maturity coefficient	83%
4	Micronaire value	3.5

Tabular form:

Three different cotton station provided their cotton for achieving the Minimum norms fixed by the mills and after lab tests in then they plot the figures in the table as mentioned below

Sr.No	Parameter	Varieties			
		X	Y	Z	Min Norms required by the mills
1	Fibre length in mm "L"	34	36	32	33 mm
2	Fibre strength "S"	24	22	19	23 Gms per tex
3	Maturity coefficient "M"	83	82	79	83%
5	Micronaire value "F"	3.4	3.8	4.10	3.5
6	Cost per Kgs "C"	160	190	140	

The above values are given as an examples only

Let

C1, C2, C3….Cn be the costs of 'n' cottons

L1, L2, L3….Ln be the lengths of each cotton variety

S1, S2, S3….Sn be the strengths of each cotton variety

M1, M2, M3….Mn be the maturity coefficients of each cotton variety

F1, F2, F3….Fn be the fineness value of each cotton grade

Then

Objective Function: Min Z = (C1*P1) + (C2*P2) + (C3*P3) + … + (Cn*Pn)

Subject to Constraints

$L1*X + L2*Y + L3*PZ \geq Lr$

$S1*X + S2*Y + S3*Z \geq Sr$

$M1*X + M2*Y + M3*Z \geq Mr$

$F1*X + F2*Y + F3*Z \leq Fr$

$X + Y + Z + = 1$

$X, Y, Z \ldots \geq 0$

Minimize Z = (190) x + (160) y + (140) Z

34 x+ 36 y + 32 z >= 33

24 x + 22 y + 19 z >= 23

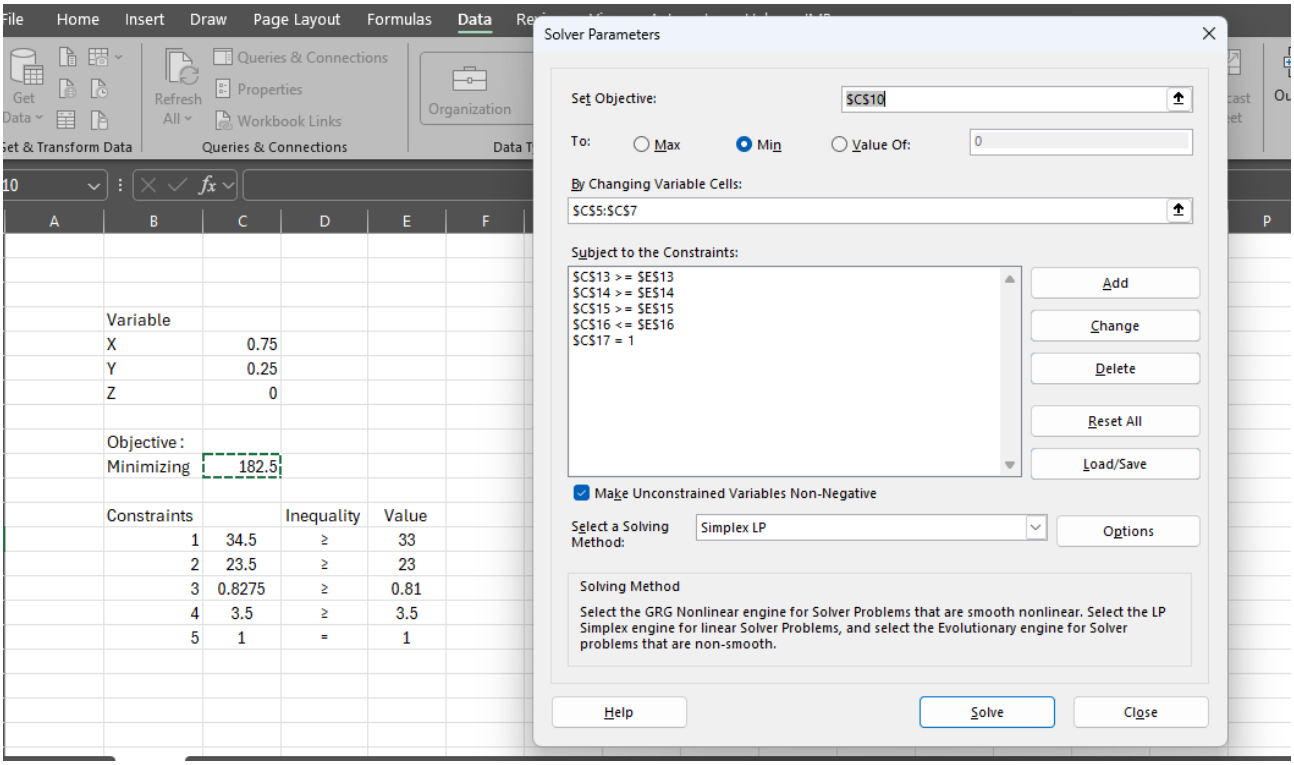
0.83 x + 0.82 y + 0.79 z >= 0.81

3.4 x + 3.8 y + 4.10 z <= 3.5

x + y + z = 1

Optimal Solution: z = 182.5; x = 0.75, y = 0.25, z = 0 this was derived by using simplex method The mill can mix Grade "X" and grade "Y" cotton varieties in 75:25 ratio so that average cost of cotton per kg will be 182.50 Rs yet they can achieve the minimum norms required for making 90s Ne .

We can do this in excel https://youtu.be/viObCm_AQv0



Example # 2:

To achieve maximum profit what must be the optimum ratio for a textile mill to process both Karded and combed yarns

The total production capacity of a spinning mills is 4000 Kgs per day and they are making both Karded and combed yarns for knitting industry. Yarn realization in Karded and combed yarns are 92% and 80% respectively. If the selling price of Karded and combed yarns per kgs are 140 & 170 Rs what should be the optimum ratio for running both types of yarns for achieving maximum profit. Also mill what's to fix overall yarn realization to be 88%

Now for the above problem we have to find the variable and constraints so that it can be converted into a mathematical argument.

Let's keep "X" for Karded and "Y" for combed process,

Selling price of Karded and combed yarn S1 and S2

Equation for Maximum profit Z = X*S1 + Y*S2 subject to

Constraints are

X+Y<=4000 since the total capacity of both karded and combed yarn production cannot exceed 4000 Kgs

(0.92)X + (0.80) Y >=0.88 Mills what's to achieve Minimum 88% yarn realization overall

x>=0
y>=0

X+Y=1 as the sum of individual process can't exceed 100%

By solving the above equation by using simplex method we reach

Optimal Solution: p = 150; x = 0.666667, y = 0.333333

In order to achieve maximum profit they can produce 66.66% Karded yarn and 33.33% combed yarn (2666 kgs of karded yarn and 1333.6 Kgs of combed yarn) so that average selling price of the product will be 150 Rs / Kg yet achieving Min 88% overall yarn realization.

Suppose if they are ready to compromise the yarn realization further by 5% i.e. 82% then by going through the same process we will reach **Optimal Solution: p = 162.5; x = 0.25, y = 0.75**

In this case the maximum profit can be achieved by producing 25% Karded yarn and 75% combed yarn (1000 kgs of karded yarn and 3000 Kgs of combed yarn) so that average selling price of the product will be 162.5 Rs / Kg at 83% overall yarn realization.

Example # 3

How to maximize profit by selecting one or more counts from a given plan

Four different counts along with their process parameters are given

Sr.No	Parameter	Value			
1	Count In Ne	C1	C2	C3	C4
2	Selling price / Kg	S1	S2	S3	S4
5	YR	W1	W2	W3	W4
6	Gms per spindle	G1	G2	G3	G4
7	UKG	U1	U2	U3	U4

The constraints in that mill are as follows

1. The prevailing raw material cost per kg is around 74 Rs so they what to have overall YR Min 71 %
2. The project is relatively new and the condition of machine and machine components are good so they try to keep productivity in GPS to Min 50

3. They reply purely on EB so they like to maintain UKG below 6

Sr.No	Parameter	Value				
1	Count In Ne	40S KW	40S CW	80S CW	100s CW	Constraint
2	Selling price / Kg	172	196	430	470	
5	YR	84	70	69	69	Min 71%
6	Gms per spindle	110	115	50	35	Min 50
7	UKG	4.5	5	8	10.5	Max 6

The above values are given as an examples only

Equation for Maximum profit P can be given by P = S1.C1 + S2.C2+…+Sn.Cn subject to

MAXIMIZE P = (170) C1+ (196) C2+ (430) C3+ (470) C4 subject to

Constraints

(84)C1+ (70) C2+ (69) C3+ (69) C4 >= 71

(110)C1+ (115) C2+ (50) C3+ (35) C4 >=50

(4.5)C1+ (5) C2+ (8) C3+ (10.5) C4 <= 6

A+B+C+D=1

The above equations can be solved by using simplex method

Optimal Solution: p = 281.429; a = 0.571429, b = 0, c = 0.428571, d = 0

In order to achieve maximum profit of Rs 281 per Kg of yarn they can allocate 57% of the spindles for making 40s KW (C1) and 43% for producing 80s Cw (C3)

Example # 4

How to optimize draft at spinning machine based on its performance. Suppose a mill what's to produce 120s NE with certain quality norms and machine performance. What's must be the ideal draft to be maintained to achieve that

Sr.No	Parameter	Value			
1	Draft	D1	D2	D3	D4
2	Imperfection / Km	Q1	Q2	Q3	Q4
3	EBR	B1	B2	B3	B4
4	GPS	P1	P2	P3	P4

The constraints in that mill are as follows

In 120 Ne they what to maintain below 475 IPI with Max 6 Breaks per 100 spl hour and minimum 36 grams per spindle

Sr.No	Parameter	Value	

1	Process	A	B	C	D	Constraint
2	Draft	40	45	55	65	
3	Imperfection / Km	400	450	480	490	Max 475
4	EBR	5.5	6	6.5	7	Max 6
5	GPS	30	35	40	44	Min 36

The above values are given as an examples only

Equation for Optimized draft D can be given by D = 40.A + 45.B + 55.C + 65D subject to

Constraints

(400)A+ (450) B+ (480) C+ (490) D <= 475

(5.5)A+ (6) B+ (6.5) C+ (7) D <=6

(30)A+ (35) B+ (40) C+ (44) D >=36

The above equations can be solved by using simplex method . Increasing or decreasing the draft in ring frame somehow affects the yarn quality parameter, performance of the machine thus affecting productivity. So in some sense the relation between those are linear. For example increasing the draft may increase the productivity but decrease the yarn quality and affects EBR. So draft is a function of productivity but subject to constraints like poor yarn quality and increased EBR

Optimal Solution: d = 55.7143; a = 0, b = 0, c = 0, d = 0.857143

If they maintain 2.154 Ne roving hank it will give an optimum draft of 55.71 thus maintaining the constraints as per mills requirement.

Example # 5

A textile retail unit wants to liquidate 300 pair of its shirts and 200 pairs of pants made from last season. They have decided to put together two offers, A and B. Offer A is a package of one shirt and one pant which will sell for 1800 RS. Offer B is a package of three shirts and one pants, which will sell for 2550 RS. The mill does not want to sell less than 30 packages of Offer A and less than 15 of Offer B. How many packages of each do they have to sell to maximize the money generated from the promotion?

Now converting the above into mathematical argument

x = number of packages of Offer A

y = number of packages of Offer B

Equation for Maximum profit can be given as P = 1800x + 2550y

	A	B	Minimal
Shirts	1	3	300
Pants	1	1	200

Constraints

$x + 3y \leq 300$

$x + y \leq 200$

$X \geq 30$

$Y \geq 15$

If we solve the equation by using simplex method we get Optimal Solution: p = 397500; x = 150, y = 50

So if they sell offer A package 150 nos which comprise of 1 shirt and 1 pant and offer B package 50 nos which has of 3 shirt and 1 pant they will make maximize the revenue generation.

Example # 6

A mill has 1 lakh spindles capacity to make compact, fancy yarn, slub and normal yarn. However, management has only 25 lakh rupees totally to spend and compact conversion costs 35 lakh to mill, fancy yarn conversion costs 25 lakh to Mill, slub yarn conversion cost 20 lakh rupees and normal yarn cost just 6 lakh rupees for spares change. If the profit is 3 lakh rupees due to compact yarn, 1.80 lakh due to fancy yarn , 1.2 lakh rupees in slub and 0.2 lakh in normal yarn how many spindles of each should be converted to maximize profits?

Now we should convert the above statements into mathematical argument

Variable	Type of yarn	Conversion cost	Profit after conversion
A	Compact	35 lakh	3 lakh
B	Fancy	25 lakh	1.80 lakh
C	Slub	20 lakh	1.20 lakh
D	Normal	6 lakh	0.2 lakh

Equation for maximizing profit P = (3) A+ (1.80) B+ (1.20) C+0.2(D) subject to constraints

(35) A+ (25) B+ (20) C+ (6) D <=25

A+B+C+D=1

If we solve the equation by using simplex method the p = 2.03448; a = 0.655172, b = 0, c = 0, d = 0.344828

So for the allocated budget of 25 lakh rupees maximum profit can be made by converting 65.5% into compact and rest for producing normal yarn. The Maximum profit will be 2.03 lakh

Suppose if budget was reduced to 17 lakh then p = 1.26207; a = 0.37931, b = 0, c = 0, d = 0.62069. It means 37,931 spindles into compact and 62,069 into normal yarn.

Example # 7

A textile company owns two units. Mill "A" costs 20,000 Rs per day to operate, and it can produce 400 Kgs of finer count yarn, 300 Kgs of Karded yarn, and 200 kgs of slub yarn per day. Mill "B" is newer and more modern. It costs 25,000 per day to operate, and it can produce 450 Kgs of finer count yarn, 390 Kgs of Karded yarn, and 270 kgs of slub yarn per day

The company has orders totaling 25,000 Kgs of finer count yarn, 27,000 Kgs of Karded yarn, and 30,000 Kgs of slub yarn. How many days should it run each Mill to minimize its costs and still make enough yarn to meet its orders?

Variable	Mill A	Mill A
Operation cost / Day	20,000	25,000
Finer count yarn	400	300
Karded yarn	300	400
Slub Yarn	200	500

Equation for minimized cost C = (20,000) A + (25,000) B subject to following constraints

Product	Production / Day	Production / Day	Constraint
Finer count yarn	400	300	(400) A + 300 (B) >=25,000
Karded yarn	300	400	(300) A + 400 (B) >=27,000
Slub Yarn	200	500	(200) A + 500 (B) >=30,000

Also A >=0, B>=0

If we solve the above equations by using simplex method we get optimum solution for minimized cost Optimal Solution: c = 1750000; a = 25, b = 50

If they operate Mill "A" for 25 days and Mill "B" for 50 days they can meet out the order.

To cross check just substitute the value of A & B in below equations

Constraint	After substituting	
(400) A + 300 (B) >=25,000	(400) 25 + 300 (50) >=25,000	10,000+15000 = 25,000
(300) A + 400 (B) >=27,000	(300) 25 + 400 (50) >=27,000	7500+20,000=27,500
(200) A + 500 (B) >=30,000	(200) 25 + 500 (50) >=30,000	5000+25000= 30,000

In fact they have 500 kgs surplus production of Karded yarn.

Suppose the demand for finer count yarn and Karded yarn was increased to 36000 Kgs & 33000 Kgs then for making it the new production target for finer count yarn as 36,000 kgs then (400) A + 300 (B) >=36,000. (300) A + 400 (B) >=33,000

By solving we will know that Mill A must be operated for 64.28 Days and Mill B for 34.28 Days for minimized cost.

Constraint	After substituting	
(400) A + 300 (B) >=36,000	(400) 64.28 + 300 (34.28) >=36,000	25,712+10284 = 35,996
(300) A + 400 (B) >=33,000	(300) 64.28 + 400 (34.28) >=33,000	19284 + 13712=32,996
(200) A + 500 (B) >=30,000	(200) 64.28 + 500 (34.28) >=30,000	12856+17140= 29, 996

Example # 8

Fibre quality Vs Yarn quality – A linear approach

We all know that the ultimate quality of yarn is decided by fibre quality, speeds and process parameters maintained at various stages of spinning machinery and its quality.

Some important fibre variety and its average quality parameters

Cotton	SL mm	UR %	MIC mg/in	BS g/tex	Cost / Kg
A	25.2	48	5.1	20.7	65
B	31.2	48	3.5	23.6	79
C	33.1	49	3.2	23.7	94

A mill what's to make 64s Ne having the below mentioned quality norms that's goes for weaving

Parameters	Norms
Count	64s Ne
Cv m%	13.8
Thin -40%	118 / Km
Thin -50%	6 / Km
Thick +35%	312 / km
Thick +50%	36 / km
Neps +140%	452 / km
Neps +200%	83 / km

In order to meet out the yarn quality norms it was observed that the fiber properties should be as mentioned below

SL mm	UR %	MIC mg/in	BS g/tex
31.00 to 32.00	48 and above	3.8 to 4.5	22 to 23

How to minimize cost by choosing the right type of fiber or fibers

P = (65) A + (79) B + (94) C subject to constraints

(25.2) A + (31.2) B + (33.1) C >=31

(48) A + (48) B + (49) C >=48

(5.1) A + (3.5) B + (3.2) C >=3.8 <=4.5

(20.7) A + (23.6) B + (23.7) C >=22 <=23

A>=0

B>=0

C>=0

A+B+C=1

Optimal Solution: p = 78.7503; a = 0.0398341, b = 0.939637, c = 0.0205289

It means Cotton type B is more suitable for making that count

Example # 09

Optimizing the drafting apron life for optimum performance machine WRT aprons cuts , Depression , Winding cuts and End Break / 100 Spl Hour :

A spinning mill wants to optimize the life of drafting pair of aprons WRT aprons cuts , Depression , Winding cuts and End Break / 100 Spl Hour :

Usage period in months	6	9	12	15	Constraints
Apron cuts / 1000 Spl / day	0.1	0.25	0.40	0.55	<=0.30
EBR	3	4.7	5.6	7	<=5
Winding cuts / 100 Km	60	80	90	120	<=90
Depression in mm	0.04	0.05	0.055	0.067	<=0.05

Optimize life of drafting apron P can be given by the equation

P = (6) A + (9) B + (12) + (15) D subject to following constraints

0.1A + 0.25B + 0.40C + 0.55D <=0.3

0.04A + 0.05B + 0.055C + 0.067D <=0.05

60A + 80B + 90C + 120D <=90

3a + 4.7b + 5.6c +7d <=5

Optimal Solution: p = 10; a = 0.333333, b = 0, c = 0.666667, d = 0 this was derived by using simplex method

It means maintaining 10 Months life in drafting pair of aprons will results in EBR ≤5, Winding cuts ≤ 90, Depression ≤ 0.05 mm and apron cuts / 1000 Spl / Day <=0.30

Conclusion: The application of linear programming in the textile spinning industry provides a systematic approach to optimize decision-making processes, such as raw material selection, product mix, and operational settings. By converting complex production scenarios into mathematical models, LP allows for the identification of optimal solutions that enhance profitability and efficiency while adhering to quality standards.

Published on 06.04.2015

04-09-2024

X

Sujai Balasubramaniam
Deputy General Manager
Signed by: 3f1b25d5-eec9-42b8-9c14-5b5b9d19d52e

Reference:

- Classes of extremal problems]. *Doklady Akad Sci SSSR*. **28**: 211–214.
- F. L. Hitchcock: *The distribution of a product from several sources to numerous localities*, Journal of Mathematics and Physics, 20, 1941, 224–230.
- G.B Dantzig: *Maximization of a linear function of variables subject to linear inequalities*, 1947. Published pp. 339–347 in T.C. Koopmans (ed.):*Activity Analysis of Production and Allocation*, New York-London 1951 (Wiley & Chapman-Hall)
- J. E. Beasley, editor. *Advances in Linear and Integer Programming*. Oxford Science, 1996. (Collection of surveys)
- *Bland, Robert G. (1977). "New Finite Pivoting Rules for the Simplex Method". Mathematics of Operations Research. **2** (2): 103–107. doi:10.1287/moor.2.2.103. JSTOR 3689647.*
- Karl-Heinz Borgwardt, *The Simplex Algorithm: A Probabilistic Analysis*, Algorithms and Combinatorics, Volume 1, Springer-Verlag, 1987. (Average behavior on random problems)
- Richard W. Cottle, ed. *The Basic George B. Dantzig*. Stanford Business Books, Stanford University Press, Stanford, California, 2003. (Selected papers by George B. Dantzig)
- George B. Dantzig and Mukund N. Thapa. 1997. *Linear programming 1: Introduction*. Springer-Verlag.
- George B. Dantzig and Mukund N. Thapa. 2003. *Linear Programming 2: Theory and Extensions*. Springer-Verlag. (Comprehensive, covering e.g. pivoting and interior-point algorithms, large-scale problems, decomposition following Dantzig–Wolfe and Benders, and introducing stochastic programming.)
- *Edmonds, Jack; Giles, Rick (1977). "A Min-Max Relation for Submodular Functions on Graphs". Studies in Integer Programming. Annals of Discrete Mathematics. **1**. pp. 185–204. doi:10.1016/S0167-5060(08)70734-9. ISBN 978-0-7204-0765-5.*
- *Fukuda, Komei; Terlaky, Tamás (1997). Thomas M. Liebling; Dominique de Werra (eds.). "Criss-cross methods: A fresh view on pivot algorithms". Mathematical Programming, Series B. **79** (1–3): 369–395. CiteSeerX 10.1.1.36.9373. doi:10.1007/BF02614325. MR 1464775. S2CID 2794181.*
- *Gondzio, Jacek; Terlaky, Tamás (1996). "3 A computational view of interior point methods". In J. E. Beasley (ed.). Advances in linear and integer programming. Oxford Lecture Series in Mathematics and its Applications. **4**. New York: Oxford University Press. pp. 103–144. MR 1438311. Postscript file at website of Gondzio and at McMaster University website of Terlaky.*
- *Murty, Katta G. (1983). Linear programming. New York: John Wiley & Sons, Inc. pp. xix+482. ISBN 978-0-471-09725-9. MR 0720547. (comprehensive reference to classical approaches).*
- Evar D. Nering and Albert W. Tucker, 1993, *Linear Programs and Related Problems*, Academic Press. (elementary)
- M. Padberg, *Linear Optimization and Extensions*, Second Edition, Springer-Verlag, 1999. (Carefully written account of primal and dual simplex algorithms and projective algorithms, with an introduction to integer linear programming – featuring the traveling salesman problem for Odysseus.)
- Christos H. Papadimitriou and Kenneth Steiglitz, *Combinatorial Optimization: Algorithms and Complexity*, Corrected republication with a new preface, Dover. (computer science)
- *Michael J. Todd (February 2002). "The many facets of linear programming". Mathematical Programming. **91**(3): 417–436. doi:10.1007/s101070100261. S2CID 6464735.* (Invited survey, from the International Symposium on Mathematical Programming.)
- *Vanderbei, Robert J. (2001). Linear Programming: Foundations and Extensions. Springer Verlag.*
- *Vazirani, Vijay V. (2001). Approximation Algorithms. Springer-Verlag. ISBN 978-3-540-65367-7.* (Computer science)

Further reading

- Dmitris Alevras and Manfred W. Padberg, *Linear Optimization and Extensions: Problems and Solutions*, Universitext, Springer-Verlag, 2001. (Problems from Padberg with solutions.)
- *Mark de Berg, Marc van Kreveld, Mark Overmars, and Otfried Schwarzkopf (2000). Computational Geometry (2nd revised ed.). Springer-Verlag. ISBN 978-3-540-65620-3.* Chapter 4: Linear Programming: pp. 63–94. Describes a randomized half-plane intersection algorithm for linear programming.
- *Michael R. Garey and David S. Johnson(1979). Computers and Intractability: A Guide to the Theory of NP-Completeness. W.H. Freeman. ISBN 978-0-7167-1045-5.*A6: MP1: INTEGER PROGRAMMING, pg.245. (computer science, complexity theory)
- *Gärtner, Bernd; Matoušek, Jiří (2006). Understanding and Using Linear Programming. Berlin: Springer. ISBN 3-540-30697-8.* (elementary introduction for mathematicians and computer scientists)
- Cornelis Roos, Tamás Terlaky, Jean-Philippe Vial, *Interior Point Methods for Linear Optimization*, Second Edition, Springer-Verlag, 2006. (Graduate level)
- *Alexander Schrijver (2003). Combinatorial optimization: polyhedra and efficiency. Springer.*
- Alexander Schrijver, *Theory of Linear and Integer Programming*. John Wiley & sons, 1998, ISBN 0-471-98232-6 (mathematical)
- *Gerard Sierksma; Yori Zwols (2015). Linear and Integer Optimization: Theory and Practice. CRC Press. ISBN 978-1-498-71016-9.*
- *Gerard Sierksma; Diptesh Ghosh (2010). Networks in Action; Text and Computer Exercises in Network Optimization. Springer. ISBN 978-1-4419-5512-8.* (linear optimization modeling)
- H. P. Williams, *Model Building in Mathematical Programming*, Fifth Edition, 2013. (Modeling)
- Stephen J. Wright, 1997, *Primal-Dual Interior-Point Methods*, SIAM. (Graduate level)
- Yinyu Ye, 1997, *Interior Point Algorithms: Theory and Analysis*, Wiley. (Advanced graduate-level)
- Ziegler, Günter M., Chapters 1–3 and 6–7 in *Lectures on Polytopes*, Springer-Verlag, New York, 1994. (Geometry)

View publication stats